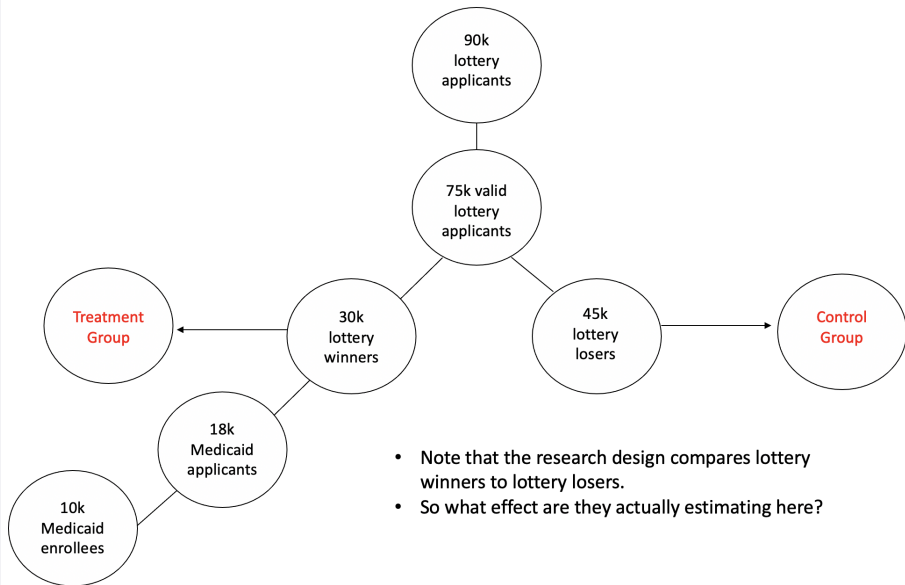


Policy Evaluation - Natural Experiments & the Oregon Medicaid Study Pt.2

April 9, 2026

Oregon Health Study



- Intent-to-treat effect (ITT): The effect of winning the lottery on health.
 - ▶ But only 10k of the 30k lottery winners actually enrolled in Medicaid.
 - ▶ Randomization does not imply compliance.
 - ▶ So the ITT estimate should *understate* the true effect of gaining Medicaid coverage on health.
- Why didn't the researchers focus on the effect of gaining Medicaid coverage instead of the ITT effect?
- Treatment-on-the-treated effect (TOT): The effect of gaining Medicaid coverage on health.
 - Divide the ITT by the share of lottery winners gaining coverage.

- Balance Test

Table A13: Lottery list characteristics and balance of treatment and controls

	Control Mean (std dev)	Difference between treatment and control		
	(1)	Full sample (2)	Credit report subsample (3)	Survey respondents subsample (4)
Panel A: Lottery list characteristics				
Year of birth	1968.0 (12.255)	0.162 (0.100)	0.136 (0.119)	-0.066 (0.191)
Female	0.557 (0.497)	-0.0069 (0.0033)	-0.0027 (0.0041)	-0.0042 (0.0068)
English as preferred language	0.922 (0.268)	0.0024 (0.0026)	0.0042 (0.0029)	-0.00033 (0.0048)
Signed up self	0.918 (0.274)	0.00030 (0.00028)	0.00060 (0.0010)	-0.0016 (0.0027)
Signed up first day of lottery	0.093 (0.290)	0.0012 (0.0025)	0.00093 (0.0031)	0.0061 (0.0049)
Gave phone number	0.862 (0.345)	-0.0029 (0.0028)	0.000088 (0.0034)	0.0059 (0.0037)
Address a PO Box	0.117 (0.321)	0.00044 (0.0027)	0.0023 (0.0034)	-0.0023 (0.0053)
In MSA	0.773 (0.419)	-0.0024 (0.0036)	-0.0018 (0.0044)	0.0011 (0.0070)
Zip code median household income	39,265.4 (8463.542)	44.891 (72.887)	12.998 (89.653)	22.031 (135.815)

- Results - Hospital Admissions

HOSPITAL UTILIZATION

	Control mean (1)	ITT (2)	LATE (3)	<i>p</i> -values (4)
Panel A: Extensive margin				
All hospital admissions	0.067 (0.250)	0.0054 (0.0019)	0.021 (0.0074)	[0.004]
Admissions through ER	0.048 (0.214)	0.0018 (0.0016)	0.0070 (0.0062)	[0.265]
Admissions not through ER	0.029 (0.167)	0.0041 (0.0013)	0.016 (0.0051)	[0.002]

- Results - Hospital Admissions

HOSPITAL UTILIZATION				
	Control mean (1)	ITT (2)	LATE (3)	<i>p</i> -values (4)
Panel B: All hospital admissions				
Days	0.498 (3.795)	0.026 (0.027)	0.101 (0.104)	[0.329] {0.328}
List charges	2,613 (19,942)	258 (146)	1,009 (569)	[0.077] {0.106}
Procedures	0.155 (1.08)	0.018 (0.0083)	0.070 (0.032)	[0.031] {0.059}

- Results - Hospital Admissions

HOSPITAL UTILIZATION				
	Control mean (1)	ITT (2)	LATE (3)	<i>p</i> -values (4)
Panel C: Admissions through ER				
Days	0.299 (2.326)	0.023 (0.017)	0.089 (0.067)	[0.183] {0.187}
List charges	1,502 (12,749)	163 (96)	636 (376)	[0.091] {0.171}
Procedures	0.081 (0.694)	0.0080 (0.0054)	0.031 (0.021)	[0.135] {0.187}

- Results - Hospital Admissions

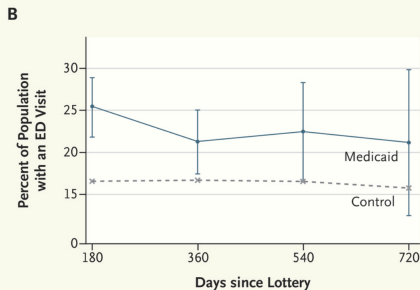
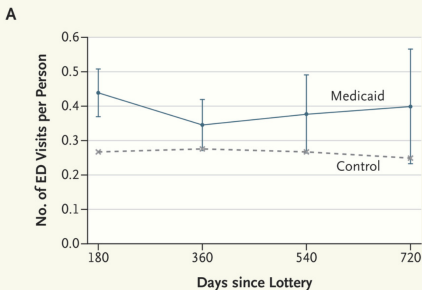
HOSPITAL UTILIZATION				
	Control mean (1)	ITT (2)	LATE (3)	<i>p</i> -values (4)
Panel D: Admissions not through ER				
Days	0.199 (2.38)	0.0033 (0.017)	0.013 (0.065)	[0.841] {0.842}
List charges	1,110 (12,422)	98 (91)	384 (356)	[0.281] {0.383}
Procedures	0.075 (0.708)	0.010 (0.0056)	0.038 (0.022)	[0.080] {0.162}

● Results - ED Visits

Perspective

Effect of Medicaid Coverage on ED Use — Further Evidence from Oregon's Experiment

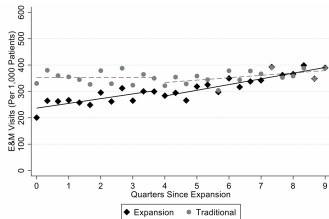
Amy N. Finkelstein, Ph.D., Sarah L. Taubman, Ph.D., Heidi L. Allen, Ph.D., Bill J. Wright, Ph.D., and Katherine Baicker, Ph.D.



Louisiana Medicaid Expansion

- Results - E&M and ED Visits

Figure 1: Evaluation and Management Visits for Expansion vs. Non-Expansion Beneficiaries



Louisiana Medicaid Expansion

● Results - E&M and ED Visits

Figure 1: Evaluation and Management Visits for Expansion vs. Non-Expansion Beneficiaries

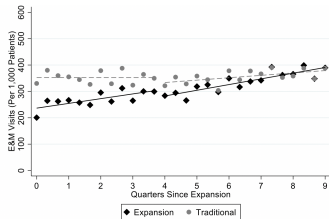
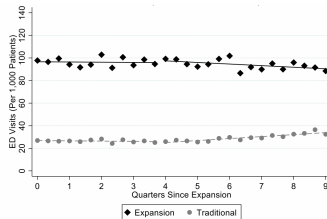


Figure 2: Emergency Department Visits for Expansion vs. Non-Expansion Beneficiaries



● Results - Preventive Care

COMPLIANCE WITH RECOMMENDED PREVENTIVE CARE (SURVEY DATA)

	Control mean (1)	ITT (2)	LATE (3)	<i>p</i> -values (4)
Blood cholesterol checked (ever)	0.625 (0.484)	0.033 (0.0074)	0.114 (0.026)	[<0.0001] {<0.0001}
Blood tested for high blood sugar/diabetes (ever)	0.604 (0.489)	0.026 (0.0074)	0.090 (0.026)	[0.0004] {<0.0001}
Mammogram within last 12 months (women ≥ 40)	0.298 (0.457)	0.055 (0.012)	0.187 (0.04)	[<0.0001] {<0.0001}
Pap test within last 12 months (women)	0.406 (0.491)	0.051 (0.01)	0.183 (0.034)	[<0.0001] {<0.0001}

- Results - Financial Strain

FINANCIAL STRAIN (ADMINISTRATIVE DATA)

	Control mean (1)	ITT (2)	LATE (3)	<i>p</i> -values (4)
Panel A: Overall				
Any bankruptcy	0.014 (0.119)	0.0022 (0.0014)	0.0086 (0.0053)	[0.106] {0.358}
Any lien	0.021 (0.144)	0.0012 (0.0014)	0.0047 (0.0056)	[0.406] {0.698}
Any judgment	0.064 (0.244)	0.0014 (0.0024)	0.0054 (0.010)	[0.573] {0.698}
Any collection	0.500 (0.500)	-0.012 (0.0041)	-0.048 (0.016)	[0.003] {0.013}
Any delinquency (credit accounts)	0.366 (0.482)	0.0016 (0.0042)	0.0063 (0.017)	[0.704] {0.698}

- Results - Financial Strain

FINANCIAL STRAIN (ADMINISTRATIVE DATA)

	Control mean (1)	ITT (2)	LATE (3)	<i>p</i> -values (4)
Panel B: Medical debt				
Any medical collection	0.281 (0.449)	-0.016 (0.0040)	-0.064 (0.016)	[<0.0001] {<0.0001}
Amount owed in medical collections	1,999 (6733)	-99 (45)	-390 (177)	[0.028] {0.025}

- Results - Financial Strain

FINANCIAL STRAIN (ADMINISTRATIVE DATA)

	Control mean (1)	ITT (2)	LATE (3)	<i>p</i> -values (4)
Panel C: Nonmedical debt				
Any nonmedical collection	0.392 (0.488)	-0.0046 (0.0041)	-0.018 (0.016)	[0.264] {0.455}
Amount owed in nonmedical collections	2,740 (9,492)	-20 (63)	-79 (248)	[0.751] {0.752}

- Results - Financial Strain

FINANCIAL STRAIN (SURVEY DATA)

	Control mean (1)	ITT (2)	LATE (3)	<i>p</i> -values (4)
Any out of pocket medical expenses, last six months	0.555 (0.497)	-0.058 (0.0077)	-0.200 (0.026)	[<0.0001] {<0.0001}
Owe money for medical expenses currently	0.597 (0.491)	-0.052 (0.0076)	-0.180 (0.026)	[<0.0001] {<0.0001}
Borrowed money or skipped other bills to pay medical bills, last six months	0.364 (0.481)	-0.045 (0.0073)	-0.154 (0.025)	[<0.0001] {<0.0001}
Refused treatment because of med- ical debt, last six months	0.081 (0.273)	-0.011 (0.0041)	-0.036 (0.014)	[0.01] {0.01}

● Results - Health

HEALTH				
	Control mean (1)	ITT (2)	LATE (3)	p-values (4)
Panel A: Administrative data				
Alive	0.992 (0.092)	0.00032 (0.00068)	0.0013 (0.0027)	[0.638]
Panel B: Survey data				
Self-reported health good/very good/excellent (not fair or poor)	0.548 (0.498)	0.039 (0.0076)	0.133 (0.026)	[<0.0001] [<0.0001]
Self-reported health not poor (fair, good, very good, or excellent)	0.86 (0.347)	0.029 (0.0051)	0.099 (0.018)	[<0.0001] [<0.0001]
Health about the same or gotten better over last six months	0.714 (0.452)	0.033 (0.0067)	0.113 (0.023)	[<0.0001] [<0.0001]
# of days physical health good, past 30 days*	21.862 (10.384)	0.381 (0.162)	1.317 (0.563)	[0.019] [0.018]
# days poor physical or mental health did not impair usual activity, past 30 days*	20.329 (10.939)	0.459 (0.175)	1.585 (0.606)	[0.009] [0.015]
# of days mental health good, past 30 days*	18.738 (11.445)	0.603 (0.184)	2.082 (0.64)	[0.001] [0.003]
Did not screen positive for depression, last two weeks	0.671 (0.470)	0.023 (0.0071)	0.078 (0.025)	[0.001] [0.003]

● Results - Mechanisms

POTENTIAL MECHANISMS FOR IMPROVED HEALTH (SURVEY DATA)

	Control mean (1)	ITT (2)	LATE (3)	<i>p</i> -values (4)
Panel A: Access to care				
Have usual place of clinic-based care	0.499 (0.500)	0.099 (0.0080)	0.339 (0.027)	[<0.0001]
Have personal doctor	0.490 (0.500)	0.081 (0.0077)	0.280 (0.026)	[<0.0001]
Got all needed medical care, last six months	0.684 (0.465)	0.069 (0.0063)	0.239 (0.022)	[<0.0001]
Got all needed drugs, last six months	0.765 (0.424)	0.056 (0.0055)	0.195 (0.019)	[<0.0001]
Didn't use ER for nonemergency, last six months	0.916 (0.278)	-0.0011 (0.0043)	-0.0037 (0.015)	[0.804]

● Results - Mechanisms

POTENTIAL MECHANISMS FOR IMPROVED HEALTH (SURVEY DATA)

	Control mean (1)	ITT (2)	LATE (3)	<i>p</i> -values (4)
Panel B: Quality of care				
Quality of care received last six months good/very good/excellent (conditional on any)	0.708 (0.455)	0.043 (0.0081)	0.142 (0.027)	[<0.0001]
Panel C: Happiness				
Very happy or pretty happy (vs. not too happy)	0.594 (0.491)	0.056 (0.0074)	0.191 (0.026)	[<0.0001]

● Results - Biometrics

Variable	Mean Value in Control Group	Change with Medicaid Coverage (95% CI) [†]	P Value
Blood pressure			
Systolic (mm Hg)	119.3±16.9	-0.52 (-2.97 to 1.93)	0.68
Diastolic (mm Hg)	76.0±12.1	-0.81 (-2.65 to 1.04)	0.39
Elevated (%) [‡]	16.3	-1.33 (-7.16 to 4.49)	0.65
Hypertension			
Diagnosis after lottery (%) ^{§¶}	5.6	1.76 (-1.89 to 5.40)	0.34
Current use of medication for hypertension (%) [§]	13.9	0.66 (-4.48 to 5.80)	0.80
Cholesterol ^{***}			
Total level (mg/dl)	204.1±34.0	2.20 (-3.44 to 7.84)	0.45
High total level (%)	14.1	-2.43 (-7.75 to 2.89)	0.37
HDL level (mg/dl)	47.6±13.1	0.83 (-1.31 to 2.98)	0.45
Low HDL level (%)	28.0	-2.82 (-10.28 to 4.64)	0.46
Hypercholesterolemia			
Diagnosis after lottery (%) ^{§¶}	6.1	2.39 (-1.52 to 6.29)	0.23
Current use of medication for high cholesterol level (%) [§]	8.5	3.80 (-0.75 to 8.35)	0.10
Glycated hemoglobin			
Level (%)	5.3±0.6	0.01 (-0.09 to 0.11)	0.82
Level ≥6.5% (%) ^{††}	5.1	-0.93 (-4.44 to 2.59)	0.61
Diabetes			
Diagnosis after lottery (%) ^{§¶}	1.1	3.83 (1.93 to 5.73)	<0.001
Current use of medication for diabetes (%) [§]	6.4	5.43 (1.39 to 9.48)	0.008

● Results - Biometrics

Variable	Mean Value in Control Group	Change with Medicaid Coverage (95% CI) [†]	P Value
Depression			
Positive screening result (%) ^{‡‡}	30.0	−9.15 (−16.70 to −1.60)	0.02
Diagnosis after lottery (%) ^{§¶}	4.8	3.81 (0.15 to 7.46)	0.04
Current use of medication for depression (%) [§]	16.8	5.49 (−0.46 to 11.45)	0.07
Framingham risk score (%) ^{§§}			
Overall	8.2±7.5	−0.21 (−1.56 to 1.15)	0.76
High-risk diagnosis	11.6±8.3	1.63 (−1.11 to 4.37)	0.24
Age of 50–64 yr	13.9±8.2	−0.37 (−2.64 to 1.90)	0.75

- Results Summary
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 - ▶ 6.7 percent of the control group experienced a hospitalization compared to 8.8 percent of the treatment group.
 - ▶ Absolute change = 2.1 *percentage points*
 - ▶ Relative change = $2.1/6.7 = 31.3$ *percent*

- Results Summary - Administrative Data:

- ▶ Hospital Care

- Hospital admissions ↑ 31.3% ($p=0.004$)
- Non-ED LOS ↑ 20% ($p=0.329$)
- Non-ED hospital charges ↑ 39% ($p=0.077$)
- Non-ED hospital procedures ↑ 45% ($p=0.031$)

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- ▶ Financial Strain

- Credit report collection ↓ 9.6% ($p=.003$)
- Medical collection ↓ 23% ($p<0.001$)
- Medical debt ↓ 20% ($p=0.028$)

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- Medical collection ↓ 23% ($p<0.001$)
- Medical debt ↓ 20% ($p=0.028$)

▶ Biometrics

- Elevated blood pressure ↓ 0.8% ($p=0.65$)
- High cholesterol ↓ 17% ($p=0.37$)
- Framingham high-risk ↑ 14% ($p=0.24$)
- Depression medication ↑ 33% ($p=0.07$)
- Positive depression screening ↓ 31% ($p=0.02$)
- Mortality ↓ 0.14% ($p=0.638$)

- Results Summary - Survey Data:

- ▶ Preventive Care

- Cholesterol check ↑ 18% ($p < 0.001$)
 - Blood glucose check ↑ 15% ($p < 0.001$)
 - Mammogram ↑ 63% ($p < 0.001$)
 - Pap test ↑ 45% ($p < 0.001$)

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- Good health ↑ 25% ($p < 0.001$)
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- Borrowed money to pay medical bills ↓ 42% ($p < 0.001$)

- ▶ Health

- Good health ↑ 25% ($p < 0.001$)
- Improved health ↑ 16% ($p < 0.001$)

- ▶ Mechanisms

- Usual source of care ↑ 68% ($p < 0.001$)
- Got all needed care ↑ 35% ($p < 0.001$)

Oregon Health Study

- The Oregon Medicaid Study found no statistically significant effects of winning the lottery on biometric measures of health. Frakt argues that we should not interpret these findings as evidence that Medicaid does not improve health. Why not?

Randomization vs. Regression

- How important is selection bias in this case?

Randomization vs. Regression

- How important is selection bias in this case?
- *“Results from the randomized evaluation showed that Medicaid substantially improved self-reported health. However if we analyzed the same data using observational methods rather than taking advantage of the randomization, results suggested that Medicaid actually worsens these same self-reported health measures.”*

- How important is selection bias in this case?

ORIGINAL ARTICLES

Primary Payer Status Affects Mortality for Major Surgical Operations

Damien J. LaPar, MD, Castigliano M. Bhamidipati, DO,* Carlos M. Mery, MD, MPH,*
George J. Stukenborg, PhD,† David R. Jones, MD,* Bruce D. Schirmer, MD,* Irving L. Kron, MD,*
and Gorav Ailawadi, MD**

Multivariable logistic regression was performed to calculate the adjusted odds of in-hospital death and in-hospital complications among patients undergoing major surgical operations. All preoperative variables entered as covariates (patient age, gender, elective operative status, mean income, hospital geographic region, teaching hospital status, type of operation, primary payer status, and categories for comorbid disease) were selected a priori based upon established clinical risk or were considered potential confounders for the effect of payer status among patients. All covariates contributing cases to each estimated outcome,

Randomization vs. Regression

	Private	Medicare	Uninsured	Medicaid
Mortality	1.00	1.54	1.74	1.97
Length of Stay	7.38	8.77	7.01	10.49
Total Costs	63,057	69,408	65,667	79,140

Medicaid Is Worse Than No Coverage at All

New research shows that patients on this government plan fare poorly. So why does the president want to shove one in four Americans into it?

Mar 2, 2011, 12:23pm EST

By Scott Gottlieb

Updated March 10, 2011 12:01 am ET

Medicaid's awful results: Column

Glenn Harlan Reynolds

Published 11:48 a.m. ET Nov. 11, 2013

CRITICAL CONDITION

Why Medicaid is a Humanitarian Catastrophe



Avik Roy Forbes Staff

The Apothecary Contributor Group

Policy

Commentary from Forbes' Policy Editor

UVa Study: Surgical Patients on Medicaid Are 13% More Likely to Die Than Those Without Insurance

By AVIK ROY | July 17, 2010 11:36 PM

